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Zhenyu Tang, Lilong Wang, Zhen Guo, Qianqian Liang, Shuyue Xue, Chengcheng Feng, Lili Ran, Lingzhi Chen, Xiaosong Wang & Jun Li

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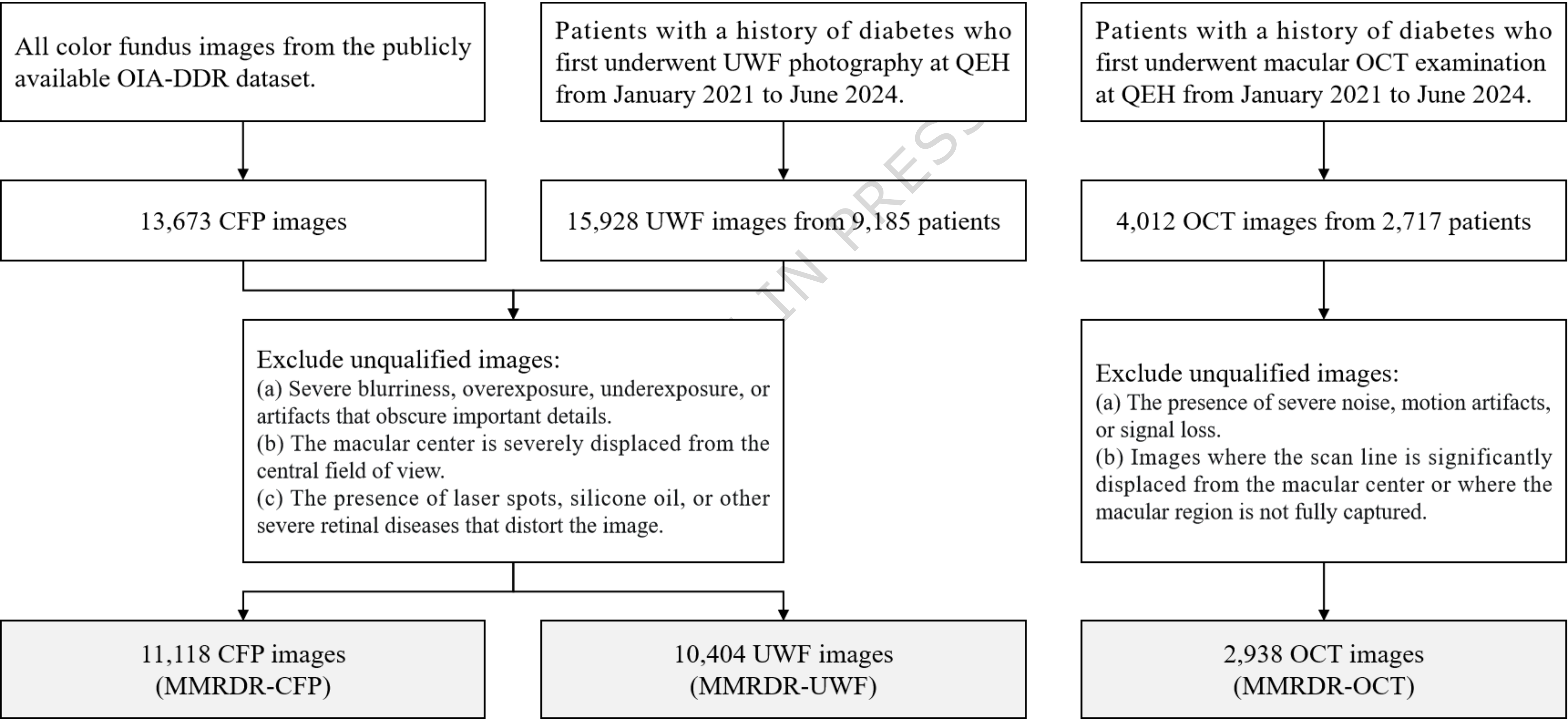
A multimodal retinal image dataset for diabetic retinopathy detection using foundation models

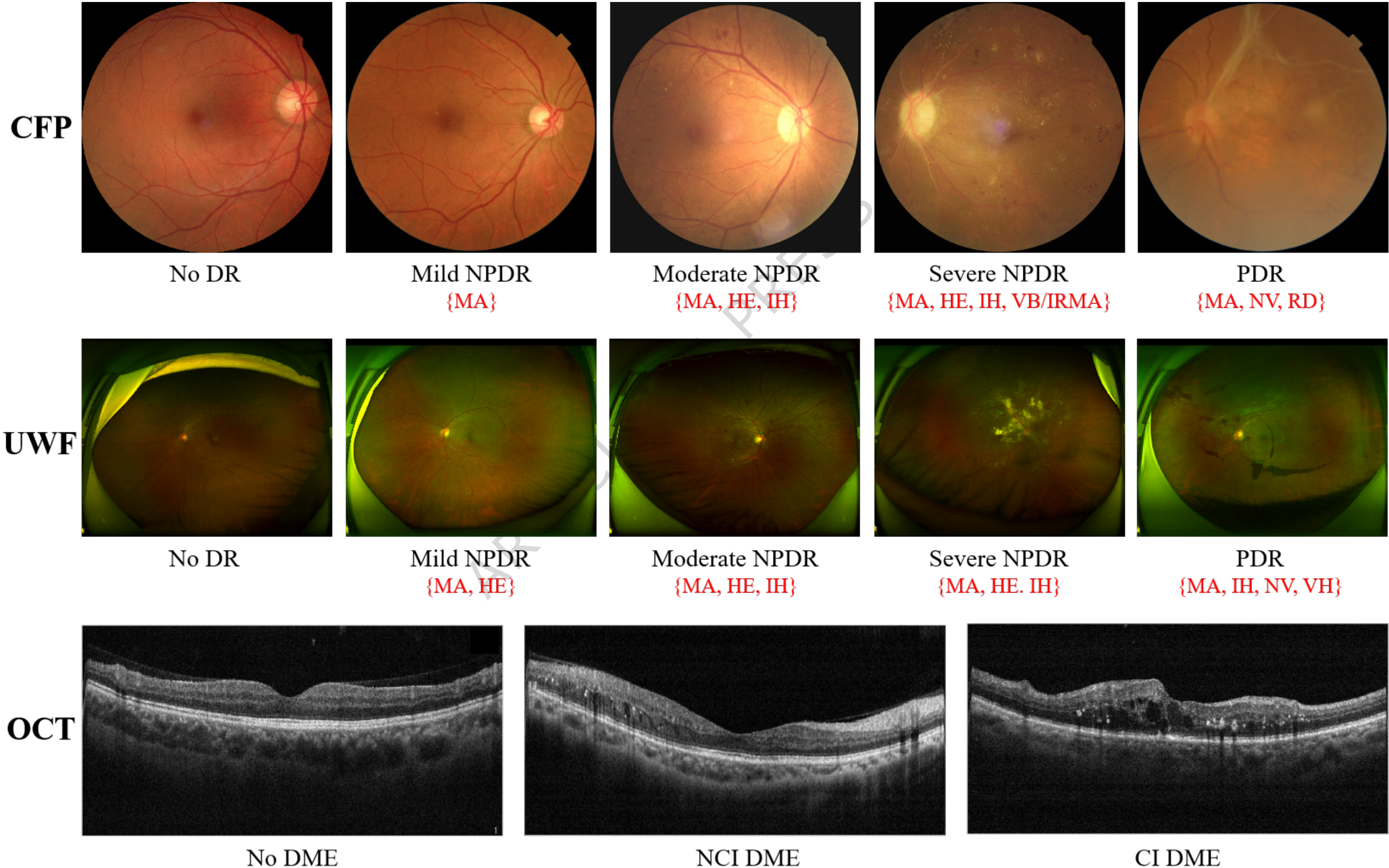
Authors: Jun Li (Qingdao Eye Hospital of Shandong First Medical Hospital), Zhenyu Tang (Shanghai Jiao Tong University), Lilong Wang (Shanghai AI Laboratory), Zhen Guo (Eye Institute of Shandong First Medical University), Qianqian Liang (Qingdao Eye Hospital of Shandong First Medical University), Shuyue Xue (Qingdao Eye Hospital of Shandong First Medical University), Chengcheng Feng (Qingdao Eye Hospital of Shandong First Medical University), Lili Ran (Qingdao Eye Hospital of Shandong First Medical University), Lingzhi Chen (Shanghai AI Laboratory), and Xiaosong Wang (Shanghai AI Laboratory)

Abstract:
Diabetic Retinopathy (DR), a leading cause of preventable blindness worldwide, underscores the urgent need for robust AI-driven diagnostic tools. Although various deep learning models for retinal imaging have emerged, their evaluation remains constrained by limited public available datasets that lack both large-scale coverage and fine-grained annotations, compromising reliable assessments of model generalizability. To bridge this gap, we introduce a comprehensive multimodal dataset that includes three key retinal imaging modalities: color fundus photography (CFP), optical coherence tomography (OCT), and ultrawide-field fundus imaging (UWF). Our dataset is unprecedented in scale and modality diversity, provides detailed lesion-level annotations and severity grades for DR and Diabetic Macular Edema (DME). We benchmark a range of fundus foundation models and large vision-language models on this dataset, revealing critical performance gaps and domain-specific challenges. By unifying large-scale multimodal data with precisely annotated clinical labels, our work establishes a foundational benchmark to drive advances in AI reliability and real-world clinical utility.

Datasets:

Repository Name	Dataset Title	Accession Number or DOI	URL to data record	Private reviewer access URL/code
MMRDR	MMRDR	10.6084/m9.figshare.29423747.v2	https://figshare.com/articles/dataset/MMRDR/29423747	


All color fundus images from the publicly available OIA-DDR dataset.
Patients with a history of diabetes who first underwent UWF photography at QEH from January 2021 to June 2024.
Patients with a history of diabetes who first underwent macular OCT examination at QEH from January 2021 to June 2024.
13,673 CFP images
15,928 UWF images from 9,185 patients
4,012 OCT images from 2,717 patients
Exclude unqualified images:
(a) Severe blurriness, overexposure, underexposure, or artifacts that obscure important details.
(b) The macular center is severely displaced from the central field of view.
(c) The presence of laser spots, silicone oil, or other severe retinal diseases that distort the image.
Exclude unqualified images:
(a) The presence of severe noise, motion artifacts, or signal loss.
(b) Images where the scan line is significantly displaced from the macular center or where the macular region is not fully captured.
11,118 CFP images
(MMRDR-CFP)
10,404 UWF images
(MMRDR-UWF)
2,938 OCT images
(MMRDR-OCT)


CFP
No DR
Mild NPDR
{MA}
Moderate NPDR
{MA, HE, IH}
Severe NPDR
{MA, HE, IH, VB/IRMA}
PDR
{MA, NV, RD}
UWF
No DR
Mild NPDR
{MA, HE}
Moderate NPDR
{MA, HE, IH}
Severe NPDR
{MA, HE. IH}
PDR
{MA, IH, NV, VH}
OCT
No DME
NCI DME
CI DME

```
MMRDR/
├── MMRDR-CFP/             # Color Fundus Photography
|   ├── img/               # CFP images (JPEG format)
|   |   ├── tr000001.jpg   # Training case 1
|   |   ├── tr000002.jpg   # Training case 2
|   |   ├── ...
|   |   ├── ts000001.jpg   # Testing case 1
|   |   └── ...
|   └── FP.csv             # Metadata & annotations
|
├── MMRDR-OCT/             # Optical Coherence Tomography
|   ├── img/               # OCT images (JPEG format)
|   |   ├── tr000001.jpg   # Training case 1
|   |   ├── ...
|   |   ├── ts000001.jpg   # Testing case 1
|   |   └── ...
|   └── OCT.csv            # Metadata & annotations
|
└── MMRDR-UWF/             # Ultra-WideField Imaging
    ├── img/               # UWF images (JPEG format)
    |   ├── tr000001.jpg   # Training case 1
    |   ├── ...
    |   ├── ts000001.jpg   # Testing case 1
    |   └── ...
    └── UWF.csv            # Metadata & annotations
```

A multimodal retinal image dataset for diabetic retinopathy detection using foundation models

Zhenyu Tang[1,2*], Lilong Wang[2*], Zhen Guo[3,4*], Qianqian Liang[4], Shuyue Xue[4], Chengcheng Feng[4], Lili Ran[4], Lingzhi Chen[2], Xiaosong Wang[2†], Jun Li[3,4†]

February 26, 2026

[1]Shanghai Jiao Tong University, Shanghai, China
[2]Shanghai AI Laboratory, Shanghai, China
[3]Eye Institute of Shandong First Medical University, Shandong, China
[4]Qingdao Eye Hospital of Shandong First Medical University, Shandong, China

∗ These authors contributed equally
† Corresponding author
Corresponding author(s): wangxiaosong@pjlab.org.cn, doctor_li@126.com

Abstract

Diabetic Retinopathy (DR), a leading cause of preventable blindness worldwide, underscores the urgent need for robust AI-driven diagnostic tools. Although various deep learning models for retinal imaging have emerged, their evaluation remains constrained by limited public available datasets that lack both large-scale coverage and fine-grained annotations, compromising reliable assessments of model generalizability. To bridge this gap, we introduce a comprehensive multimodal dataset that includes three key retinal imaging modalities: color fundus photography (CFP), optical coherence tomography (OCT), and ultrawide-field fundus imaging (UWF). Our dataset is unprecedented in scale and modality diversity, provides detailed lesion-level annotations and severity grades for DR and Diabetic Macular Edema (DME). We benchmark a range of fundus foundation models and large vision-language models on this dataset, revealing critical performance gaps and domain-specific challenges. By unifying large-scale multimodal data with precisely annotated clinical labels, our work establishes a foundational benchmark to drive advances in AI reliability and real-world clinical utility.

Background & Summary

Diabetic Retinopathy (DR) is a leading cause of preventable blindness worldwide, particularly among working-age adults [1–3]. A common and vision-threatening complication of DR is Diabetic Macular Edema (DME), caused by fluid leakage into the macula that can result in severe central vision loss [3, 4]. With the rising global prevalence of diabetes, the number of individuals affected by DR is projected to reach approximately 191 million by 2030 [5]. This highlights the urgent need for early detection, timely intervention, and ongoing eye care. Effective management depends on regular screening and accurate staging of retinopathy, as prompt treatment can significantly reduce disease progression and preserve visual function [6].

Color fundus photography (CFP) remains the primary imaging modality for DR screening and classification, particularly in primary care and community-based settings [7]. It enables non-invasive visualization of the retina and facilitates the detection of hallmark lesions such as microaneurysms, hemorrhages, exudates, and neovascularization. However, standard CFP typically captures a limited field of view (30–60°), which may fail to detect peripheral retinal abnormalities often present in early or proliferative stages of DR. In contrast, ultra-widefield fundus imaging (UWF), commonly used in large hospitals or specialized ophthalmic centers, extends coverage to approximately 200° of the retinal periphery. This enables earlier and more comprehensive identification of sight-threatening lesions that conventional CFP may miss. Several clinical studies have demonstrated that UWF imaging detects a higher prevalence of peripheral DR-related lesions compared to standard CFP [8, 9]. These findings highlight the added clinical utility of UWF imaging in improving early DR detection and accurate staging of disease severity. Optical Coherence Tomography (OCT) plays a central role in the

Dataset	Total Images	Modality			Annotation		
		CFP	UWF	OCT	DR classification	No. of lesion types	DME classification
Kaggle DR [11]	88,702	✓			Five grades	–	–
IDRiD [12]	516	✓			Five grades	4	–
EYEPACS [13]	35,126	✓			Five grades	–	–
OIA-DDR [14]	13,673	✓			Five grades	4	–
MedFMC-Retino [15]	1,392	✓			Five grades	–	–
DeepDRiD [16]	2,256	✓	✓		Five grades	–	–
UWF4DR 2024 [17]	800		✓		Binary	–	Binary
OUWFD [18]	700		✓		Multiple RDs	–	–
Retina-SLO [19]	7,943		✓		Multiple RDs	–	–
OCTMNIST [20]	109,309			✓	–	–	Multiple RDs
Retinal OCT-C8 [21]	24,000			✓	–	–	Multiple RDs
MMRDR (ours)	24,460	✓	✓	✓	Five grades	7	Three classes

Table 1: Summary of DR datasets across different image modalities (CFP, UWF, OCT). RD: retinal disease.

evaluation of DME, offering high-resolution, cross-sectional images of retinal layers. OCT allows precise quantification of retinal thickness, intraretinal fluid accumulation, and structural changes, which are treated as key indicators for both DME diagnosis and monitoring treatment response [3, 10]. These three retinal imaging modalities are complementary in the comprehensive evaluation and management of DR and DME.

Public datasets for DR research vary significantly in scale, modality coverage, and annotation depth, as summarized in Table 1. Large-scale collections such as the Kaggle DR dataset [11], EYEPACS [13], and OIA-DDR [14] focus primarily on standard CFP with five-grade DR severity annotations. However, some reports have highlighted issues such as poor image quality and inaccurate grading labels within these datasets [22–24]. Datasets like IDRiD [12] and OIA-DDR [14] provide additional annotations of 4 lesion types for several hundreds of images, missing several key lesions. Resources for UWF imaging are more limited, usually with much fewer image samples. DeepDRiD [16] uniquely offers CFP and UWF images with DR grading, whereas UWF4DR 2024 [17] and OUWFD [18] provide simpler binary classifications, and Retina-SLO [19] covers broader retinal pathologies rather than in-depth DR analysis. For OCT, substantial datasets exist—including OCTMNIST [20] and Retinal OCT-C8 [21], but these primarily support general retinal disease categorization rather than detailed DR or DME-specific phenotyping. Overall, current resources exhibit three critical limitations: modality integration is sparse; lesion annotation granularity is restricted; and DME characterization in OCT datasets remains rudimentary or absent.

Reflecting the clinical need for automated DR detection, a trend towards multi-modal AI analysis of DR is also rapidly emerging. Early DR detection systems relied on supervised CNNs (e.g., ResNet-50 [25]) or Vision Transformers (ViT) [26] trained for specific classification tasks. While effective in constrained settings, these models face scalability limitations due to their dependency on extensive labeled data and poor cross-modal adaptability. The recent emergence of foundation models has transformed this paradigm through self-supervised pretraining strategies including masked reconstruction [27, 28] and image-text contrastive learning [29]. RETFound [30] demonstrates how masked image reconstruction learns robust retinal representations from unlabeled images, while vision-language models like FLAIR [31] and KeepFIT [32] leverage image-text alignment to align visual features with clinical concepts and achieve high generalizability. Remarkably, KeepFIT [32] and RETFound [30] both incorporate multiple imaging modalities, including CFP and OCT, for better generalizability. Recent large vision-language models (LVLMs) such as InternVL3 [33] extend these capabilities to multimodal reasoning covering massive available modalities including CFP, OCT, and UWF, with medical-specific variants like HuatuoGPT-Vision [34] incorporating domain knowledge through visual instruction tuning on large-scale medical datasets. The comprehensive evaluation of these methods are also necessary. While these advances show significant promise, their comprehensive evaluation on DR analysis requires benchmarks covering more modalities and more fine-grained annotations, a need currently unmet by existing resources.

To overcome these challenges, we construct a large-scale dataset, **MMRDR**, comprising a total of 24,460 **M**ulti**M**odal **R**etinal images for in-depth **DR** analysis with expert annotations for: 1) 5-grade DR severity, 2) 7 lesion types on CFP and UWF, and 3) 3-class DME grading on OCT. Compared

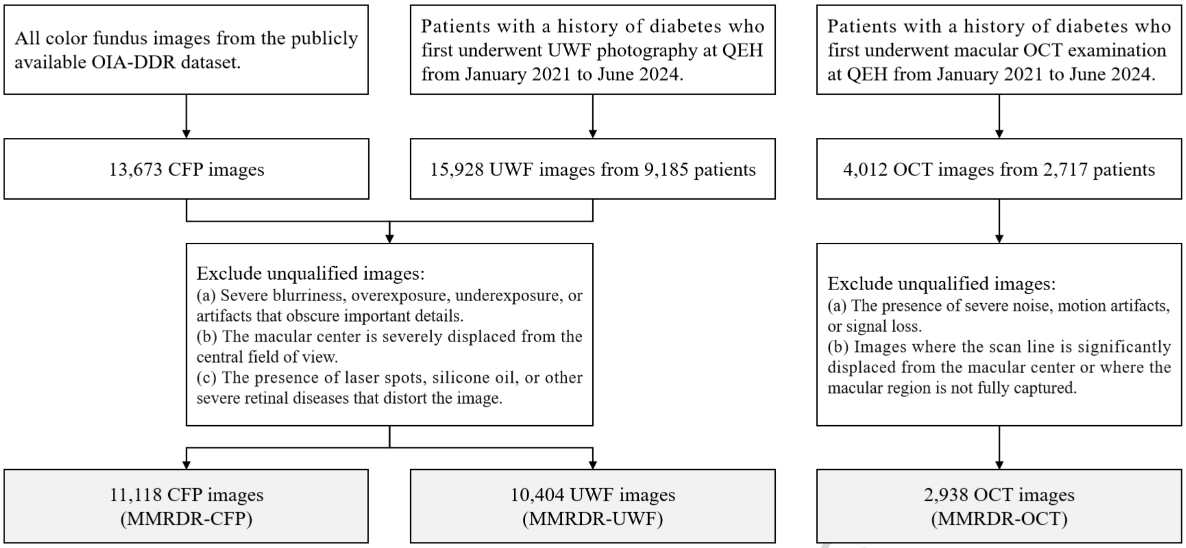


Figure 1: Illustration of data collection and image quality control for MMRDR.

with current public datasets abovementioned, MMRDR offers advantages in multimodal coverage, comprehensive grading, and lesion annotations. Rigorous data quality control and standardized labeling procedures were implemented. Using this dataset, we conduct an extensive technical validation of foundation models with diverse focuses, covering ImageNet-pretrained general vision models, ophthalmic foundation models, as well as large vision language models pretrained on either general or medical imaging domain. This dataset is designed to be a versatile resource for the research community, enabling the development and evaluation of AI models across multiple tasks, including DR grading, lesion detection, and multimodal fusion. Beyond benchmarking, MMRDR's comprehensive annotations and multimodal design make it readily reusable for advancing foundational models in ophthalmology, supporting pretraining, fine-tuning, or hybrid approaches for both specialized and generalizable AI solutions.

Methods

Ethics approval. This study was approved by the Institutional Review Board of Qingdao Eye Hospital (QEH) of Shandong First Medical University (Ref. QingYanLunShen[2024] No.41) and conducted in accordance with the Declaration of Helsinki. Informed consent was waived by the medical ethics committee due to the retrospective design.

Data collection. As shown in Figure 1, the OIA-DDR dataset, a widely used public resource for DR research, was selected as the source of CFP images for the MMRDR dataset. Both UWF and OCT images were independently obtained from the PACS system at QEH. We identified patients with a history of diabetes who had undergone their first UWF photography using the Optos UWF$^{\text{TM}}$ camera between January 2021 and June 2024. This cohort included 15,928 UWF images from 9,185 patients. Similarly, an independent cohort of 2,717 patients accepted macular OCT imaging using several devices, e.g. Zeiss Cirrus HD-OCT 5000 and Optovue RTVue-XR, with radial line scanning mode between January 2021 and June 2024. For each eye, a single representative OCT image passing through the fovea and typically exhibiting the most severe pathological lesions (e.g., intraretinal cysts, subretinal fluid, retinal detachment) was retrieved from the PACS system.

Image quality control. Two junior ophthalmologists quickly assessed the quality of the images across all three modalities and excluded any that meet the unqualified criteria listed in Figure 1. Some poor-quality examples are also demonstrated in Supplementary Figure 1. As a result, the final MMRDR dataset comprises 11,118 CFP, 10,404 UWF, and 2,938 OCT images. Detailed demographic information for MMRDR-UWF and MMRDR-OCT is provided in Table 2.

Annotation criteria. DR severity was graded based on the presence and extent of characteristic retinal lesions in accordance with established clinical criteria [3, 35]. In CFP and UWF images,

	MMRDR-CFP	MMRDR-UWF	MMRDR-OCT
Patients	–	6,109	2,005
Age (years)	–	61.65±11.06	61.79±11.08
Gender (M/F)	–	3,092/3,017	969/1,036
Eyes (L/R)	5,607/5,511	5,090/5,314	1,450/1,488
No DR	6,579	3,372	–
Mild NPDR	1,302	1,936	–
Moderate NPDR	1,959	2,166	–
Severe NPDR	603	1,377	–
PDR	675	1,553	–
Microaneurysm	4,398	6,938	–
Hard exudate	2,648	3,846	–
Intraretinal Hemorrhage	2,580	3,847	–
VB/IRMA	236	1,010	–
Neovascularization	556	1,188	–
Vitreous hemorrhage	358	1,219	–
Retinal detachment	93	238	–
No DME	–	–	1,017
NCI DME	–	–	280
CI DME	–	–	1,641

Table 2: Demographic characteristics and annotation distribution of MMRDR

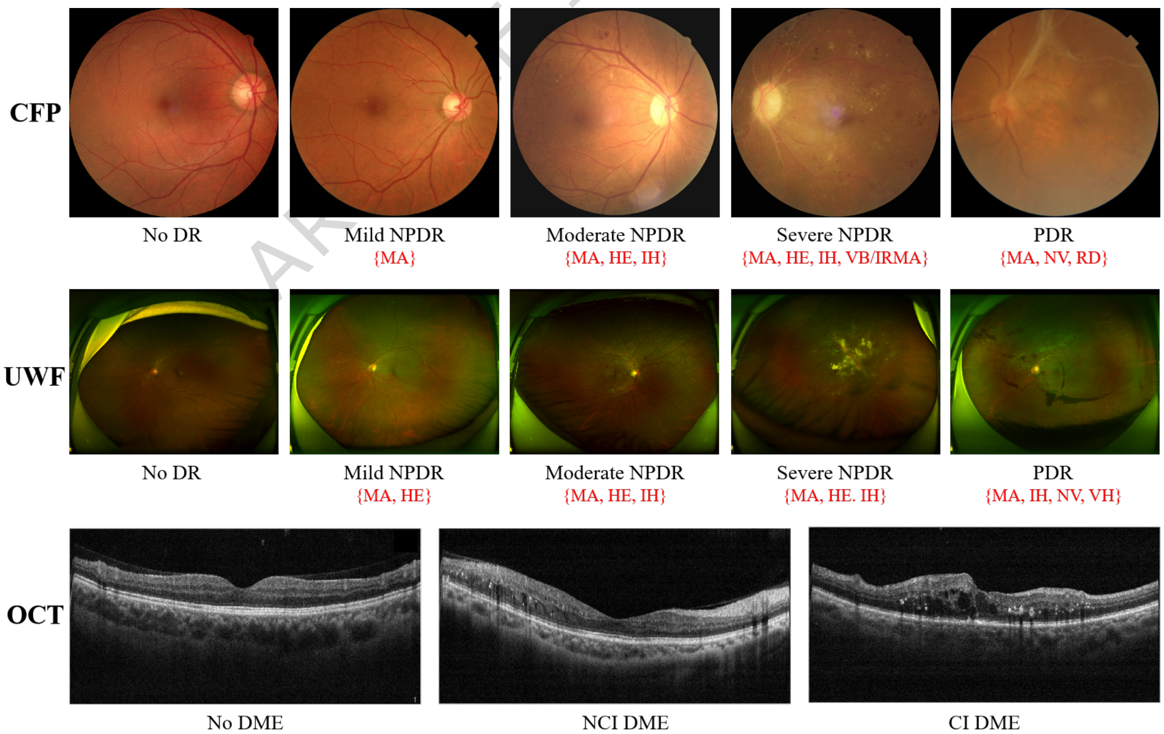


Figure 2: Example CFP, UWF, OCT images of the MMRDR dataset. The corresponding DR and DME labels are provided below each image, with DR-related lesions highlighted in red text within parentheses. MA: microaneurysm, HE: hard exudate, IH: intraretinal hemorrhage, VB/IRMA: venous beading/intraretinal microvascular abnormality, NV: neovascularization, VH: vitreous hemorrhage, RD: retinal detachment.

```
MMRDR/
├── MMRDR-CFP/              # Color Fundus Photography
│   ├── img/                # CFP images (JPEG format)
│   │   ├── tr000001.jpg    # Training case 1
│   │   ├── tr000002.jpg    # Training case 2
│   │   ├── ...
│   │   ├── ts000001.jpg    # Testing case 1
│   │   └── ...
│   └── FP.csv              # Metadata & annotations
│
├── MMRDR-OCT/              # Optical Coherence Tomography
│   ├── img/                # OCT images (JPEG format)
│   │   ├── tr000001.jpg    # Training case 1
│   │   ├── ...
│   │   ├── ts000001.jpg    # Testing case 1
│   │   └── ...
│   └── OCT.csv             # Metadata & annotations
│
└── MMRDR-UWF/              # Ultra-WideField Imaging
    ├── img/                # UWF images (JPEG format)
    │   ├── tr000001.jpg    # Training case 1
    │   ├── ...
    │   ├── ts000001.jpg    # Testing case 1
    │   └── ...
    └── UWF.csv             # Metadata & annotations
```

Figure 3: Folder structure of the MMRDR dataset.

absence of any retinal lesions was classified as no DR. Mild NPDR was defined by the presence of microaneurysms (MA) only. Moderate NPDR was characterized by more extensive MA, intraretinal hemorrhages (IH), or hard exudates (HE) in at least one quadrant, or venous beading (VB) in a single quadrant. Severe NPDR required either severe IH or MA in all four quadrants, definite VB in two or more quadrants, or intraretinal microvascular abnormalities (IRMA) in one or more quadrants—the so-called "4-2-1 rule". Proliferative DR (PDR) was diagnosed when NV, vitreous hemorrhage (VH), or retinal detachment (RD) was observed. The seven key DR-related lesions, including MA, HE, IH, VB/IRMA, NV, VH, and RD, were systematically recorded whenever present.

For DME, grading was performed using OCT images to assess structural changes in the macula. Cases were categorized as no DME, non-center-involving DME (NCI DME), or center-involving DME (CI DME) [3, 35]. Absence of retinal thickening or intraretinal fluid indicated no DME. NCI DME was defined by the presence of retinal thickening or fluid outside the central subfield. CI DME was assigned when fluid accumulation or retinal thickening involved the foveal center, typically accompanied by increased central macular thickness exceeding normal thresholds.

Labeling process. The annotation of the MMRDR dataset was performed by four senior ophthalmologists (each with over five years of clinical experience) and one retinal specialist with more than 15 years of expertise in retinal image interpretation, all from QEH. To ensure annotation consistency and reliability, a structured training and calibration protocol was implemented. The retinal specialist first trained the ophthalmologists on the annotation criteria described in the previous section. The retinal specialist constructed a reference subset of 130 images (50 CFP, 50 UWF, 30 OCT) selected to approximately evenly represent the diagnostic categories in the MMRDR dataset. The specialist then carefully annotated this subset. Each ophthalmologist then independently labeled the same images. Inter-rater agreement was evaluated using Cohen's kappa. All four achieved $\kappa \geq 0.75$ for DR grading (CFP and UWF) and $\kappa \geq 0.80$ for DME staging (OCT), indicating substantial to high level of agreement. After calibration, the ophthalmologists independently annotated approximately 6,000 randomly assigned images. When encountering cases with diagnostic uncertainty, they referred them to the retinal specialist for final adjudication. Example CFP, UWF, and OCT images with annotations are shown in Figure 2.

Privacy. To protect patient privacy, all personally identifiable and demographic-sensitive information has been removed or replaced. Patient names and original IDs have been substituted with randomly generated identifiers. Dates of birth, scan dates, and other time-related data have also been omitted.

Data Records

The MMRDR dataset is available at figshare [36]. As shown in Figure 3, the dataset is organized into three imaging modality subsets ("MMRDR-CFP", "MMRDR-OCT", or "MMRDR-UWF"), each partitioned into training and testing sets. Within each modality's subdirectory, the image data are stored in an "img" folder in JPEG format, accompanied by a corresponding CSV annotation file ("FP.csv", "OCT.csv", or "UWF.csv", respectively). The images for each modality are partitioned randomly into training and testing subsets at an approximate 8:2 ratio. Specifically, the OCT and UWF subsets were split at the patient level to prevent data leakage. In contrast, the CFP subset was split at the image level due to the unavailability of patient identifiers in the source OIA-DDR dataset. The images adhere to a standardized naming convention: training images are named "tr[6-digit ID].jpg" and testing images "ts[6-digit ID].jpg", where the 6-digit ID ranges from 1 to the total number of images in the subset. The CSV files for the CFP and UWF modalities include a single grade value (0-4) indicating the diabetic retinopathy severity level and a multi-label lesion field represented as a 7-element binary array where each value (0 or 1) signifies the absence or presence of a specific retinal lesion type in a predefined order (the same as in Table 2). The OCT modality's CSV file contains only a single grade value (0-2) for diabetic macular edema classification. All three CSV files share the "lr" field (0 = left eye, 1 = right eye) as a laterality identifier. An exemplar code snippet is provided below on how to read a CSV annotation file and display an image:

```python
import os
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.image as mpimg

mmrdr_folder = "MMRDR" # Modify this to your local folder
uwf_folder = os.path.join(mmrdr_folder, "MMRDR-UWF")

# Load the UWF annotation file
df = pd.read_csv(os.path.join(uwf_folder, "UWF.csv"))

# Iterate over the UWF subset
for _, row in df.iterrows():
    print(f"""
Imaging modality: {row["type"]}
Image path: {row["image"]}
Grade annotation: {row["grade"]}
Lesion annotation: {row["lesion"]}
Is right eye?: {row["lr"]}
""")

    # Read and display the image
    img = mpimg.imread(os.path.join(uwf_folder, row["image"]))

    plt.figure(figsize=(8, 6))
    plt.imshow(img)
    plt.title(f"UWF Image: {row['image']}\nGrade: {row['grade']}, Lesion: {row['lesion']}")
    plt.axis('off')  # Hide axes
    plt.show()

    break # showing only 1 example here
```

Technical Validation

Baselines. In this section, We test three categories of foundation models on the proposed MMRDR dataset: 1) Large Vision Language Models (LVLMs), including general-purpose models (InternVL3 [33] variants) and the medical-specialized HuatuoGPT-Vision [34], assessed in zero-shot and fine-tuning settings; 2) ImageNet-pretrained general vision models (ResNet-50 [25] and ViT [26]); and 3) Ophthalmic foundation models (RETFound [30], FLAIR [31], KeepFIT [32]), evaluated under recommended configurations (fine-tuning the full model or with linear probing). For LVLMs, the largest variant is used for zero-shot evaluation. Due to computational constraints, we showcase LoRA [37]-based fine-tuning only for InternVL3-38B and compare it with its zero-shot performance.

Method	Setting	CFP				UWF				OCT	
		$\mathrm{Acc_G}$	$\mathrm{F1_G}$	Acc_L	$\mathrm{F1}_L$	$\mathrm{Acc_G}$	$\mathrm{F1_G}$	Acc_L	$\mathrm{F1}_L$	$\mathrm{ACC_C}$	$\mathrm{F1_C}$
Large Vision Language Models											
InternVL3-38B [33]	ZS	0.615	0.285	0.544	0.294	0.364	0.278	0.220	0.341	0.543	0.261
InternVL3-78B [33]	ZS	0.625	0.356	0.550	0.249	0.371	0.280	0.257	0.323	0.183	0.165
HuatuoGPT-Vision-34B [34]	ZS	0.634	0.433	0.517	0.230	0.453	0.413	0.152	0.239	0.215	0.240
InternVL3-38B [33]	FT*	0.781	0.659	0.705	0.651	0.598	0.611	0.436	0.714	0.861	0.700
ImageNet-pretrained General Vision Models											
ViT [26]	FT	0.805	0.694	0.940	0.682	0.690	0.676	0.910	0.766	0.883	0.700
ResNet-50 [25]	FT	0.823	0.702	0.944	0.671	0.745	0.733	0.923	0.774	0.890	0.701
Ophthalmic Foundation Models											
KeepFIT [32]	FT**	0.787	0.684	0.929	0.643	0.662	0.660	0.883	0.716	0.778	0.664
FLAIR [31]	FT**	0.806	0.707	0.922	0.673	0.666	0.659	0.870	0.685	0.728	0.559
RETFound [30]	FT	0.822	0.714	0.951	0.682	0.736	0.712	0.920	0.769	0.897	0.759

Table 3: Performance of foundation models on MMRDR-CFP. Acc: accuracy. $(\cdot)_G$: DR grading. $(\cdot)_L$: DR lesion classification. $(\cdot)_C$: DME classification. FT: fine-tune. ZS: zero-shot. *: with LoRA. **: linear probe.

Data processing and evaluation. LVLMs utilized their native image processors, while other models resized inputs to 512×512 resolution to preserve diagnostic details. Classification performance is assessed using Accuracy, and F1 scores. AUC metrics are excluded due to prohibitive computational costs in extracting logits from sequential LVLM outputs (complexity $O(n^L)$, where $n \approx 100,000$ is the token vocabulary size and $L = 1,024$ is the maximum token length).

Baseline results. Performance across modalities is presented in Table 3.

Results of pre-trained and fine-tuned models. Despite the strong general vision and language abilities, pre-trained LVLMs exhibit relatively low classification performance (Setting ZS in Table 3). After fine-tuning one of these models, InternVL3-38B, on the MMRDR training sets (Setting FT*), the performance improved significantly. This result shows that MMRDR dataset may benefit both the DR grading and lesion classification tasks across the three modalities. Moreover, ophthalmic foundation models (KeepFIT, FLAIR) demonstrated strong generalizability on MMRDR test set with minimal tuning (Setting FT** in Table 3), aligning with their pre-trained on ophthalmic data.

Results of different modalities. UWF imaging presents the greatest diagnostic challenge, with accuracy dropping by more than 7% compared to CFP across all model categories. This performance gap may stem from UWF's wider field-of-view, which captures complex pathological interactions or peripheral artifacts not present in CFP. Ophthalmic foundation models demonstrate robust cross-modal generalization regarding lesion detection performance, exhibiting an average accuracy drop (difference in $\mathrm{Acc_G}$ in Table 3, denoted as ΔAcc here) of $\leq 5.2\%$ when adapting to UWF. In contrast, LVLMs show significant degradation, with ΔAcc ranging from 29.3% to 36.5%. This suggests that transferring lesion-specific features learned from CFP to UWF is beneficial, likely because the two modalities share similar local textures and structures related to lesions, as the primary difference lies in the enlarged field-of-view of UWF. The ImageNet-pretrained ResNet-50 generalizes slightly better to UWF than ophthalmic foundation models do. This counter-intuitive result is likely attributable to limitations in the scale and annotation granularity of currently available UWF datasets for pretraining ophthalmic foundation models, underscoring the necessity of developing foundation models for multimodal retinal images including UWF.

Usage Notes

All interested researchers are strongly encouraged to download and use the proposed MMRDR dataset for their experiments and model development. It can be used to train DR detection algorithms using CFP, UWF, and OCT images, and to evaluate algorithmic performance against our established benchmarks or human annotators. We anticipate that this dataset will help foster the development of more advanced AI models for DR in-depth analysis, particularly those based on state-of-the-art foundation models.

Data Availability

The MMRDR dataset in this Data Descriptor is publicly available on figshare (https://figshare.com/articles/dataset/MMRDR/29423747) and is cited as reference [36]. The dataset comprises three distinct subsets (color fundus photography, optical coherence tomography, and ultra-widefield fundus imaging) based on imaging modality. Each subset is partitioned into training and testing sets, with image data in JPEG format and corresponding CSV files providing annotations. The annotations include diabetic retinopathy severity grades, multi-label lesion classifications for CFP and UWF, diabetic macular edema grades for OCT, and laterality identifiers.

Code Availability

The code used to evaluate the foundation models in this paper is available at GitHub (https://github.com/Vladimirovich2019/MMRDR_Evaluation).

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Author Contributions

Z.G., X.W., L.W. and J.L. conceptualized the study and compiled the dataset, while also developing the annotation protocols. Z.T. and L.C. carried out technical validation. Z.T. and L.W. drafted the majority of the manuscript. Q.L., S.X., C.F., L.R., and J.L. contributed to dataset curation and annotation. Z.G., X.W., and J.L. also provided critical scientific insights and contributed to the writing of the manuscript. All authors reviewed the final version and approved its submission.

Competing Interests

The authors declare no competing interests.